

Final Feature Definitions

Churn Prediction Model — CRM Scope

1. Purpose of This Document

This document lists the final, agreed set of features used by the churn prediction model. It is meant to remove any ambiguity about what each feature means and how it should be calculated, so the same logic is used consistently if the model is rebuilt or moved into production.

2. Feature List

The model uses 20 features in total, covering patient profile, enrollment behaviour, support interaction history, and eligibility.

Feature	Type	Definition	Calculation Logic
Age	Number	Patient's age in years.	Taken directly from patient record.
Gender	Category	Patient's gender.	Taken directly from patient record.
Region	Category	Geographic region the patient lives in.	Taken directly from patient record.
Insurance Type	Category	Type of insurance coverage.	Taken directly from patient record.
Disease Condition	Category	Medical condition being treated.	Taken directly from patient record.
Baseline Risk	Number (0-1)	Clinical risk score at treatment start.	Taken directly from patient record.
Num Programs Enrolled	Number	Distinct support programs the patient has joined.	Count of unique Program IDs per patient.
Num Enrollments	Number	Total enrollment records for the patient.	Count of enrollment rows per patient.
Num Withdrawn	Number	Times the patient withdrew from a program.	Count of enrollments with status = Withdrawn.
Enrollment Channel	Category	How the patient first joined a program.	Channel value from the patient's earliest enrollment.
Enrollment Reason	Category	Reason the patient enrolled.	Reason value from the patient's earliest enrollment.
Total Interactions	Number	Total support interactions the patient has had.	Count of interaction records per patient.
Pct Follow Up Required	Percent (0-1)	Share of interactions needing follow-up.	Average of the Follow-Up-Required flag.
Pct Resolved	Percent (0-1)	Share of interactions successfully resolved.	Share of interactions with outcome = Resolved.
Pct No Response	Percent (0-1)	Share of interactions with no patient response.	Share of interactions with outcome = No Response.
Pct Escalated	Percent (0-1)	Share of interactions that were escalated.	Share of interactions with outcome = Escalated.
Num Financial Assist Interactions	Number	Interactions related to financial assistance.	Count of interactions of that type.
Num Adherence Counseling	Number	Adherence counseling sessions received.	Count of interactions of that type.
Num Programs Eligible	Number	Programs the patient is eligible to join.	Count of unique eligible Program IDs.
Pct Enrollment Eligible	Percent (0-1)	Share of eligible programs the patient can enroll in.	Average of the Enrollment-Eligible flag.

3. Notes

- All percentage-type features are decimals between 0 and 1, not whole numbers (0.75, not 75).
- Category-type features must be encoded using the same encoder saved with the model, so new category values should be flagged and reviewed before scoring.
- Features that could leak the outcome directly (such as PDC, Persistence Days, or Discontinuation Date) were intentionally excluded from this list.